

Technical Requirement Document

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD):

- TR-DLT-001:
- Related Functional Requirement(s): FR-DLT-001
- Objective: Implement a data pipeline to load customer and order data from CSV files into Delta tables.
- Target Cluster Configuration:
- Cluster Name: Data_Load_Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2 x Standard_DS3_v2
- Autoscaling: Enabled (min: 2, max: 5)
- Auto Termination: Enabled (30 minutes)
- Libraries Installed: delta-lake, spark-csv
- Source Data Details:
- Customer CSV data: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata`` CSV
- Order CSV data: ``/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata`` CSV
- Target Data Details:
- Customer Delta table: ``customer`` Delta
- Order Delta table: ``order`` Delta
- Job Flow / Pipeline Stages:
- Read customer CSV data into a Delta table named "customer".
- Read order CSV data into a Delta table named "order".
- Data Transformations / Business Logic:
- Step: Read CSV data into Delta tables.
- Transformation Logic: Use Spark CSV reader to read CSV data and write to Delta tables.
- Error Handling and Logging:
- Log errors to a log file.
- Handle null or empty input data.
- TR-DLT-002:
- Related Functional Requirement(s): FR-DLT-002

- Objective: Add a new column "TotalAmount" to the order table.
- Target Cluster Configuration:
 - Cluster Name: Data_Transform_Cluster
 - Databricks Runtime Version: 10.4.x-scala2.12
 - Node Type: Standard_DS3_v2
 - Driver Node: 1 x Standard_DS3_v2
 - Worker Nodes: 2 x Standard_DS3_v2
 - Autoscaling: Enabled (min: 2, max: 5)
 - Auto Termination: Enabled (30 minutes)
 - Libraries Installed: delta-lake
- Source Data Details:
 - Order Delta table: `order` Delta
- Target Data Details:
 - Transformed Order Delta table: `order` Delta
- Job Flow / Pipeline Stages:
 - Add a new column "TotalAmount" to the "order" table.
 - Data Transformations / Business Logic:
 - Step: Calculate "TotalAmount".
 - Transformation Logic: Multiply "PricePerUnit" and "Qty" columns.
 - Error Handling and Logging:
 - Log errors to a log file.
 - Handle null or empty input data.
- TR-DLT-003:
 - Related Functional Requirement(s): FR-DLT-003
 - Objective: Remove null and duplicate records from customer and order tables.
 - Target Cluster Configuration:
 - Cluster Name: Data_Cleansing_Cluster
 - Databricks Runtime Version: 10.4.x-scala2.12
 - Node Type: Standard_DS3_v2
 - Driver Node: 1 x Standard_DS3_v2
 - Worker Nodes: 2 x Standard_DS3_v2
 - Autoscaling: Enabled (min: 2, max: 5)
 - Auto Termination: Enabled (30 minutes)
 - Libraries Installed: delta-lake

- Source Data Details:
- Customer Delta table: `customer` Delta
- Order Delta table: `order` Delta
- Target Data Details:
- Cleansed Customer Delta table: `customer` Delta
- Cleansed Order Delta table: `order` Delta
- Job Flow / Pipeline Stages:
- Remove rows with null values from both "customer" and "order" tables.
- Remove duplicate records from both "customer" and "order" tables.
- Data Transformations / Business Logic:
- Step: Remove null and duplicate records.
- Transformation Logic: Use Spark DataFrame API to filter and drop duplicates.
- Error Handling and Logging:
- Log errors to a log file.
- Handle null or empty input data.
- TR-DLT-004:
- Related Functional Requirement(s): FR-DLT-004
- Objective: Create an "ordersummary" table and load joined data from customer and order tables.
- Target Cluster Configuration:
- Cluster Name: Data_Load_Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2 x Standard_DS3_v2
- Autoscaling: Enabled (min: 2, max: 5)
- Auto Termination: Enabled (30 minutes)
- Libraries Installed: delta-lake
- Source Data Details:
- Customer Delta table: `customer` Delta
- Order Delta table: `order` Delta
- Target Data Details:
- ordersummary Delta table: `ordersummary` Delta
- Job Flow / Pipeline Stages:
- Join the "customer" and "order" tables on the "CustId" field.

- Load the joined data into the "ordersummary" table using SCD type 2.
- Data Transformations / Business Logic:
- Step: Join customer and order data.
- Transformation Logic: Use Spark DataFrame API to join tables.
- Error Handling and Logging:
- Log errors to a log file.
- Handle null or empty input data.
- TR-DLT-005:
- Related Functional Requirement(s): FR-DLT-005
- Objective: Update the "ordersummary" table on customer data change.
- Target Cluster Configuration:
- Cluster Name: Data_Update_Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2 x Standard_DS3_v2
- Autoscaling: Enabled (min: 2, max: 5)
- Auto Termination: Enabled (30 minutes)
- Libraries Installed: delta-lake
- Source Data Details:
- Customer Delta table: `customer` Delta
- ordersummary Delta table: `ordersummary` Delta
- Target Data Details:
- Updated ordersummary Delta table: `ordersummary` Delta
- Job Flow / Pipeline Stages:
- Detect changes in the "customer" table.
- Update the "ordersummary" table by making old records inactive and new records active.
- Data Transformations / Business Logic:
- Step: Update ordersummary table.
- Transformation Logic: Use Spark DataFrame API to update table.
- Error Handling and Logging:
- Log errors to a log file.
- Handle null or empty input data.
- TR-DLT-006:

- Related Functional Requirement(s): FR-DLT-006
- Objective: Create a "customeraggregatespend" table and load aggregated data.
- Target Cluster Configuration:
- Cluster Name: Data_Aggregate_Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2 x Standard_DS3_v2
- Autoscaling: Enabled (min: 2, max: 5)
- Auto Termination: Enabled (30 minutes)
- Libraries Installed: delta-lake
- Source Data Details:
- ordersummary Delta table: `ordersummary` Delta
- Target Data Details:
- customeraggregatespend Delta table: `customeraggregatespend` Delta
- Job Flow / Pipeline Stages:
- Aggregate the "TotalAmount" column from the "ordersummary" table by "Name" and "Date" columns.
- Load the aggregated data into the "customeraggregatespend" table.
- Data Transformations / Business Logic:
- Step: Aggregate data.
- Transformation Logic: Use Spark DataFrame API to aggregate data.
- Error Handling and Logging:
- Log errors to a log file.
- Handle null or empty input data.
- TR-DLT-007:
- Related Functional Requirement(s): FR-DLT-007
- Objective: Implement data pipelines using Delta Live Tables.
- Target Cluster Configuration:
- Cluster Name: DLT_Cluster
- Databricks Runtime Version: 10.4.x-scala2.12
- Node Type: Standard_DS3_v2
- Driver Node: 1 x Standard_DS3_v2
- Worker Nodes: 2 x Standard_DS3_v2
- Autoscaling: Enabled (min: 2, max: 5)

- Auto Termination: Enabled (30 minutes)
- Libraries Installed: delta-live-tables
- Job Flow / Pipeline Stages:
- Implement the data pipelines using Delta Live Tables to load and transform data.
- Data Transformations / Business Logic:
- Step: Implement DLT pipeline.
- Transformation Logic: Use Delta Live Tables API to implement pipeline.
- Error Handling and Logging:
- Log errors to a log file.
- Handle null or empty input data.